

MMF 1921

Operations Research

Course Outline and Requirements

May 9, 2026

Instructional Personnel

- **Instructor:** Roy H. Kwon (rkwon@mie.utoronto.ca)
- **TA:** Charlie (Xuanze) Li (xuanze.li@mail.utoronto.ca)

Lectures and Tutorials

- **Lectures (held in MMF classroom room on 17th floor OPG building)) :**
- **May 19 to May 27 10:30AM to 1:30 PM**
- **Tutorials (will take place in the MMF classroom):**
- **Tutorials will be daily from 2:30-3:30PM.**

Class Resources

- **Reference Text:** *Introduction to Linear Optimization and Extensions with MATLAB*, by Roy H. Kwon, CRC Press, Sept. 2013.
- **Class Website:** Quercus
- Slides and other material will be posted on Quercus.

Course Content

- This course will focus on mathematical optimization models and methods useful for quantitative finance. In particular, linear, robust, scenario-based, non-linear, and integer programming models and their solutions techniques.
- We will also study along major classes of financial models.

What is Operations Research (OR)?

- **Operations research (OR)** is best defined as the discipline that uses **mathematical modeling, optimization, statistics, and analytical methods** to support **better decision-making in complex systems**.
- The core idea is to turn messy, real-world decision problems into structured mathematical models that can be analyzed and optimized.
- **In MMF 1921 we focus on optimization.**

Typical Steps in Using OR

1. Determine the problem to be solved
2. Observe the system and gather data
3. Formulate a mathematical model of the problem and any important subproblems
4. Verify the model and use the model for prediction or analysis
5. Select a suitable alternative
6. Present the results to the organization
7. Implement and evaluate

Mathematical Optimization

Unconstrained Optimization

$$\textit{maximize } f(x) \textit{ where } x \in R^n$$

Constrained Optimization

$$\textit{maximize } f(x) \textit{ where } x \in S \subseteq R^n$$

Constrained Optimization

- This class of problems can be written as

minimize $f(x)$

subject to

$g_i(x) \leq 0$ for $i = 1, \dots, n$

$h_i(x) = 0$ for $i = 1, \dots, m$

- i.e. we can write
- $S = \{x \in R^n \mid g_i(x) \leq 0 \text{ for } i = 1, \dots, n, h_i(x) = 0 \text{ for } i = 1, \dots, m\}$

Optimization is indispensable in quant finance

Finance is fundamentally about **choosing actions under uncertainty**.
Optimization provides the mathematical machinery to:

- allocate capital
- control risk
- calibrate models
- design trading/execution strategies
- manage constraints
- make decisions systematically
- It is the bridge between **models** and **real-world decisions**.

Optimization at Citadel

Citadel's Global Quantitative Strategies (GQS) uses **centralized portfolio optimization** to run sophisticated strategies efficiently. This includes:

- Allocating capital across thousands of securities
- Enforcing risk limits and factor constraints
- Balancing expected returns vs. volatility
- Managing exposures across equities, futures, fixed income, and FX
- Coordinating positions across multiple pods (independent investment team) and strategies

Derivatives Pricing and Model Calibration

Firms like Citadel trade options, convertibles, rates derivatives, and structured products. Optimization is used to:

- Calibrate volatility surfaces
- Fit yield curves
- Estimate model parameters (e.g., stochastic volatility)
- Minimize pricing errors vs. market quotes

Citadel's origins in convertible bond arbitrage relied heavily on model-based pricing and optimization.

Machine Learning and Predictive Modelling

Optimization is used in:

- Training ML models (regularization, loss minimization)
- Hyperparameter tuning
- Feature selection
- Forecast blending
- **Optimization is one of the major pillars in the field of Machine Learning**

Linear Programming (LP)

Situation:

$$\text{maximize } f(x) \text{ where } x \in S \subseteq R^n$$

*All functions are **linear**.*

- Applications: Short term financing, bond portfolio construction, MAD portfolio optimization
- Theory and Algorithms: Geometry, simplex method, duality theory

LP Example

- minimize or maximize a linear objective
- subject to linear equalities and inequalities

maximize $3x + 4y$

subject to $5x + 8y \leq 24$

$x, y \geq 0$

A **feasible solution** satisfies all of the constraints.

$x = 1, y = 1$ is feasible; $x = 1, y = 3$ is **infeasible**.

An **optimal solution** is the best feasible solution.

The optimal solution is $x = 4.8, y = 0$.

Terminology

- Decision variables: e.g., x and y .
 - In general, there are quantities you can control to improve your objective which should completely describe the set of decisions to be made.
- Constraints: e.g., $5x + 8y \leq 24$, $x \geq 0$, $y \geq 0$
 - Limitations on the values of the decision variables.
- Objective Function. e.g., $3x + 4y$
 - Value measure used to rank alternatives
 - Seek to maximize or minimize this objective
 - examples: maximize NPV, minimize cost

Non-Linear Programming (Optimization) NLP

Situation:

$$\text{maximize } f(x) \text{ where } x \in S \subseteq R^n$$

*At least one function is **non – linear**.*

- Applications: Mean-variance optimization and its many variants and extensions (factor modeling approach, Black-Litterman). A brief tutorial on modern portfolio theory.
- Solution techniques: KKT conditions and iterative methods (steepest descent and gradient descent, Newton's method, interior point methods).
- Convex optimization.
- Risk-parity optimization. Robust portfolio optimization. Second-order cone programming and robust mean-variance optimization).

NLP Example

maximize $f(x,y) = xy$

subject to $x - y^2/2 \leq 10$

$3x - 4y \geq 2$

$x \geq 0, y \geq 0$

Integer Programming (Optimization) IP

- Situation:

$$\text{maximize } f(x) \text{ where } x \in S \subseteq R^n$$

At least one variable is required to be an integer value.

- Applications: Cardinality restricted mean-variance, index fund construction.
- Solution techniques: Linear programming-based branch and bound.

IP Example

- Maximize $3x_1 + 4x_2 - 3x_3$
 $3x_1 + 2x_2 - x_3 \geq 17$
 $3x_2 - x_3 = 14$
 $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$ and
 x_1, x_2, x_3 are all integers

Scenario Optimization

- Scenario: Incorporates uncertainty in parameter values in optimization model through scenarios.
- Stochastic programming.
- Applications: Risk management, CVaR optimization

Example

You want to choose a single decision variable:

$$w = \text{fraction of wealth invested in a risky asset}$$

The rest $1 - w$ goes into cash.

You face **three possible market scenarios**, each with a probability and a return Y

Scenario	Probability	Asset Return
Boom	0.3	+20%
Normal	0.5	+5%
Crash	0.2	−15%

Example continued

- A simple **scenario-based optimization** is:

$$\max_w 0.3(0.20w) + 0.5(0.05w) + 0.2(-0.15w)$$

subject to a **scenario constraint**:

$$-0.15w \geq -0.05$$

- This constraint says:
- *In the crash scenario, the loss cannot exceed 5%.*
- Solve the constraint:
- $-0.15w \geq -0.05 \Rightarrow w \leq \frac{1}{3}$
- So you can invest **at most 33.3%** in the risky asset.

Course Requirements: Homework

- There will be 3 homework assignments.
- The first homework is due May 21 (submit through quercus by 11:59PM).
- The second homework is due Monday May 25th (submit through quercus by 11:59PM).
- The third homework is NOT to be turned in.
- Only a subset of problems from each homework assignment will be marked but solutions to most problems will be provided.
- Homework assignments will be dispatched as soon as possible.

Course Requirements: Computational Assignments

- There will be two computational assignments.
- The first computational assignment will be due Sunday May 24th by 11:59PM (submit through quercus).
- The second computational assignment will be due (before 1PM on quercus) on June 1st (submit through quercus)
- More details about how to submit will be given later by the TA.
- (NOTE: the schedule might be subject to change so make sure to be aware of any such changes.)

Course Requirements: Exams

- Exam: There will be a three hour exam on May 29th from 10:30AM-1:30PM. I will give the rules for the exam later.
- All exams will take place in the MMF classroom on the 17th floor in the OPG building on 700 University Avenue.

Course Mark Breakdown

- HW#1 is worth 5%
- HW#2 is worth 5%
- HW#3 not marked
- Computational Assignment 1 is worth 21.5%
- Computational Assignment 2 is worth 25%
- Exam is worth 40%
- Class participation is 3.5%. Everybody will get this credit if attendance is not a problem. If there is less than 20 students present at a lecture I will take attendance and those not present (unless there is a valid excuse e.g. medical or you made arrangements with me ahead of time) will lose a half a percent for each class missed.